Sequences and series

1 Describe the difference between a sequence and a series.

2 Find the third term in the sequence $4, -5, 6, -7, 8$.

3 Find the next number in the sequence $16, 36, 64, 100, 144, \ldots$

4 Consider the sequence a_n. If the first term is a_1, write the notation used for the fifth term of the sequence.

5 Determine whether each sequence is finite or infinite:

a $1, 2, 3, 4, 5, 6$

b $-1, -2, -3, -4, -5, -6, \ldots$

c $-1, -2, -3, -4, -5$

d $1, 2, 3, 4, 5, 6, \ldots$

6 If all the dates in July were used to form a sequence, would the sequence be finite or infinite?

7 Identify the following sequences as finite or infinite:

a $a_1 = 2$; $a_n = 2a_{n-1}$ for $n \geq 2$

b $a_2 = 3$; $a_n = a_{n-1} + a_{n-2}$ for $3 \leq n \leq 9$

8 State the first five terms of the following sequences. Round terms to four decimal places where necessary.

a $a_n = 3n - 3$

b $a_n = -2^n$

c $a_n = 2^{n+1}$

d $a_n = (-2)^n (3n)$

e $a_n = \dfrac{1}{\sqrt{n}}$

f $a_n = \left(1 + \dfrac{1}{n}\right)^n$

9 State the first four terms, in exact form, of the following sequences:

a $a_n = \dfrac{n+4}{n+3}$

b $a_n = \dfrac{5n-1}{n^2+5}$

c $a_n = \left(\dfrac{1}{2}\right)^n (n-3)$

d $a_n = (-1)^{n-3} (n-1)$

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10 For each of the following sequences, state the first four terms:

a $a_1 = 4,\ a_n = na_{n-1} - 5$ for $n > 1$

b $a_1 = 3,\ a_n = a_{n-1} + n$ for $n > 1$

c $a_1 = -2,\ a_n = 2n + 3a_{n-1}$ for $n > 1$

d $a_1 = 3,\ a_2 = 5,\ a_n = a_{n-1} \times a_{n-2}$ for $n > 2$

e $a_1 = 3,\ a_2 = 2,\ a_n = 3\left(a_{n-1}\right)^2 - 4a_{n-2}$ for $n > 2$

Sigma notation

11 Write the following series using sigma notation:

a $4 + 8 + 12 + 16 + 20 + \ldots$

b $-3 + 9 - 27 + 81 - 243 + 729$

c $\dfrac{1}{1^3} + \dfrac{1}{2^3} + \dfrac{1}{3^3} + \dfrac{1}{4^3} + \dfrac{1}{5^3}$

d $\dfrac{1}{3 \times 1} + \dfrac{1}{3 \times 2} + \dfrac{1}{3 \times 3} + \ldots + \dfrac{1}{3 \times 7}$

e $\dfrac{2}{5+1} + \dfrac{2}{5+2} + \dfrac{2}{5+3} + \ldots + \dfrac{2}{5+11}$

f $1 - \dfrac{1}{2} + \dfrac{1}{4} - \dfrac{1}{8} + \ldots + \dfrac{1}{1024}$

12 Find the value of the following:

a $\displaystyle\sum_{r=1}^{5} (4r - 4)$

b $\displaystyle\sum_{k=3}^{5} \frac{1}{k}$

c $\displaystyle\sum_{r=1}^{4} \frac{1}{r+2}$

d $\displaystyle\sum_{k=1}^{5} (-1)^{k-1} k^2$

e $\displaystyle\sum_{r=-2}^{1} \left(3r^2 - 1\right)$

f $\displaystyle\sum_{n=1}^{5} 5\,(3)^n$

g $\displaystyle\sum_{r=1}^{5} \left(3^r + 1\right)$

h $\displaystyle\sum_{k=0}^{3} \left((-3)^k + 1\right)$

i $\displaystyle\sum_{n=1}^{5} n^{n-3}$

j $\displaystyle\sum_{n=-3}^{0} \left(n^2 - 4n\right)$

13 Given that $x_1 = 2,\ x_2 = 3,\ x_3 = 4,\ x_4 = 5$ and $x_5 = 6$, find the value of $\sum_{n=1}^{5} x_n$.

14 Given that $x_1 = 1,\ x_2 = 0,\ x_3 = -1,\ x_4 = -2$ and $x_5 = -3$, find the value of $\sum_{j=1}^{5} 2x_j$.

15 Given that $x_1 = -1,\ x_2 = 0,\ x_3 = 1,\ x_4 = 2$ and $x_5 = 3$, find the value of $\sum_{r=1}^{5} (4x_r + 2)$.

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17 Given that $x_1 = 2$, $x_2 = 1$, $x_3 = 0$, $x_4 = -1$ and $x_5 = -2$, find the value of the following:

a $\sum_{i=1}^{4}(x_i^2 - 3)$

b $\sum_{n=2}^{4}\left(\frac{x_n - 4}{x_n + 2}\right)$

18 Consider $f(x) = 4x + 1$, $x_1 = -1$, $x_2 = -2$, $x_3 = -3$, $x_4 = -4$ and $\delta x = \frac{3}{2}$.

a $\sum_{k=1}^{4} f(x_k)\,\delta x$

b $\sum_{x=1}^{5} f(x)$

19 Consider the series $\sum_{i=1}^{7}\left(i + i^3\right)$ and $\sum_{i=1}^{7} i + \sum_{i=1}^{7} i^3$.

a Write each series in expanded form.

b Do the two sums have the same value?

c Is $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$?

Applications

20 A doctor administered pain medication in the following amounts in milligrams over four days after their patient underwent surgery:

Year (n)	1	2	3	4
Dosage (m)	2000	1520	1155.2	878.0

Find the recursive formula which models this sequence. Round the values to two decimal places if necessary.

21 Christa deposits $\$1$ into a new bank account while Jenny deposits $\$38$ into a new bank account. The next day, Christa adds $\$2$ to the account, and doubles how much she adds each day thereafter. Jenny adds $\$5$ each day.

a List the amount Christa had in her account each day for the first four days.

b List the amount Jenny had in her account each day for the first four days.

c On which day will they have the same amount of money in their bank accounts?

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